

WQU MSFE 690 CAPSTONE PROJECT - MODULE 4 DESIGN

Project Proposal - Student Group 17830

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1. Research Question

Do higher ESG disclosure scores reduce volatility and improve risk-adjusted returns for JSE mining companies (2019-2023)? H1: Higher ESG -> lower volatility (GARCH), H2: Lower CVaR, H3: Causal effect.

2. Motivation

Mining core to SA economy, high ESG scrutiny. JSE ESG guidance 2022.

3. Data

JSE_sample_data (2019-2023) - 12 firms, 60 months. Variables: Ticker, Date, Close, Return, ESG Score, MktCap. File: JSE_sample_data (1).csv

4. Methodology

- a) Panel Fixed Effects: $\text{Return} = \alpha + \beta_1 \cdot \text{ESG} + \text{Controls}$
 - b) GARCH(1,1): Volatility by ESG quartile
 - c) CVaR 95%: High vs Low ESG
 - d) Causal Bayesian Network (pgmpy)
 - e) Long High-ESG / Short Low-ESG backtest
- Code: esg_jse_mining_analysis.ipynb

5. Expected Contribution

Quantify ESG alpha for JSE mining, risk management framework, open-source repo.

6. Division of Labor

Ting Ting Han: Literature & Data

Henry Mayambu: GARCH/CVaR & GitHub Repo

Jonathan Matura: Causal Network & Backtest

7. Repo Contents

README.md, requirements.txt, data csv, ipynb, png, CONTRIBUTORS.md